

Branden Palmer

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Education

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- University of Texas at Austin** | *Austin, TX, USA* **Jan 2026 – May 2029**
- M.S. in Computer Science
- Arizona State University** | *Tempe, AZ, USA* **Aug 2020 – Dec 2025**
- B.S. in Computer Science | Summa Cum Laude
 - GPA: 3.83 | Dean's List Spring 2024, Fall 2024, Spring 2025, Fall 2025

Projects

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- Autonomous Driving Trajectory Prediction** | *University of Texas at Austin* | *Deep Learning Project* **Jan 2026 – Apr 2026**
- Python PyTorch Transformers Tensorboard CNNs CUDA NumPy Matplotlib*
- Built and trained deep learning models in PyTorch to predict future vehicle trajectories for autonomous driving from lane geometry and raw image inputs
 - Implemented and compared an MLP planner, Transformer-based planner, and CNN planner for waypoint prediction on the SuperTuxKart Drive Dataset
 - Developed model-agnostic training pipelines from scratch, including data loading, optimization, checkpointing, and offline evaluation
 - Evaluated planning accuracy using longitudinal and lateral trajectory error, comparing structured-input and end-to-end vision-based driving approaches
- Applied Stylometry** | [Video](#) | *Arizona State University* | *Data Science Report* **Nov 2025 – Dec 2025**
- Python pandas NumPy scikit-learn NLP TF-IDF PCA SciPy Matplotlib Plotly*
- Applied stylometry and NLP techniques to analyze authorship signals across multiple text corpora using TF-IDF, character n-grams, and sentence-level features
 - Built an end-to-end text analysis pipeline in Python, including HTML parsing, text normalization, feature extraction, and exploratory visualization
 - Reduced high-dimensional linguistic feature spaces using PCA and visualized clustering behavior to compare content-based vs style-based representations
 - Evaluated methodological limitations of stylometry on short-form texts, demonstrating how data volume and shared background can confound authorship attribution

Open Source Contributions

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- Job Sourcerer** | [Video](#) | [Link](#) | *A centralized job application platform* **Jan 2025 – Dec 2025**
- Drizzle ORM PGLite PostgreSQL React Shadcn TailwindCSS TypeScript WXT*
- Setup CD/CI pipeline utilizing GitHub Actions, enabling early regression detection and single click deployment to Firefox and Chrome Extension stores
 - Handcrafted client side ATS algorithm utilizing cosine of similarity, achieved resume to job description matching within 5% of paid service at zero cost
 - Implemented browser extension architecture using background, content, and popup scripts to extract job data from third-party sites (e.g., Handshake) and persist structured records locally
 - Developed a client-side data layer using Drizzle ORM with a PGLite-backed database to manage jobs, resumes, and derived analytics without a backend dependency
- G8 Medical Technology** | [Link](#) | *Medical App* **Jan 2024 – Apr 2024**
- Java JavaFX SQLite*
- Broke six-person team into three feature focused subteams, reduced crosstalk, increased dev momentum
 - Performed 37 merges, instrumental in synchronizing six-person team
 - Delivered all required specifications on time
- Super Productivity** | [Link](#) | *Productivity App* **Apr 2022 – Jan 2024**
- SCSS Angular.js*
- ID'd regression bug by utilizing git bisect
 - Corrected UI issue by isolating problematic SCSS, resulting change merged into production branch

Other Experience

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- Senior Aircraft Technician** | *Lockheed Martin, Balad AB, Iraq* **Dec 2017 – Jan 2020**
- Held active secret security clearance*